Finite Reflection Groups
and Coxeter Classification

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Humphreys — Reflection Groups and Coxeter Groups, Cambridge University Press, 1990

GOAL

Classify all finite groups generated by reflections in real Euclidean space using **Coxeter graphs.**

The key insight: classifying finite reflection groups = classifying connected positive definite Coxeter graphs.

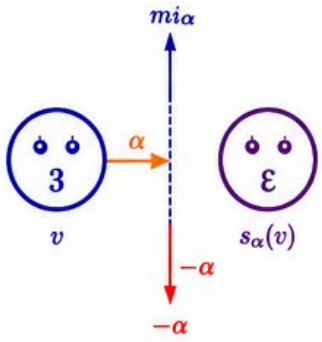
1. REFLECTION – DEFINITION

Reflection s_α in the hyperplane orthogonal to the root α is given by

$$s_\alpha(v) = v - \frac{2(v,\alpha)}{(\alpha,\alpha)}\,\alpha$$

Properties:

1. $s_\alpha^2 = id$ (order 2)
2. Orthogonal transformation of V
3. $\det(s_\alpha) = -1$



3. COXETER GROUPS

A Coxeter group W is generated by reflections $s_1, \ldots, s_n$ with relations

$$(s_i s_j)^{m_{ij}} = 1$$

where

$$m_{ii} = 1, \quad m_{ij} \geq 2$$

Finite Coxeter group
⇔ Finite reflection group

If the Coxeter graph is disconnected, W is the direct product of the groups corresponding to its connected components.

Key Fact
Coxeter graph of any finite reflection group is always positive definite.

2. ROOT SYSTEM, REFLECTION GROUP & EXAMPLE

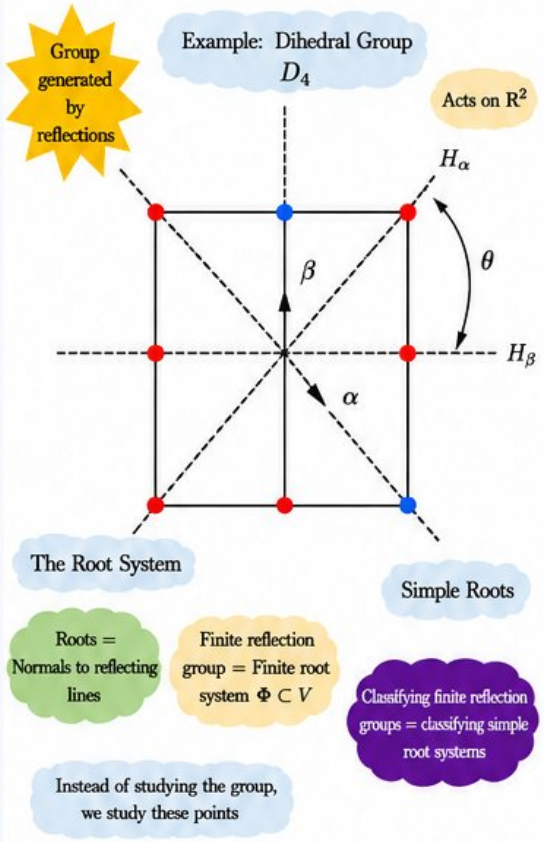


Figure 1: The case $m = 4$

A **root system** Φ (finite subset of V) satisfies:

- $\Phi = \Phi^+ \cup \Phi^-$
- Φ is closed under reflections
- Simple roots $\Delta \subset \Phi$ form a basis for span(Φ)
- $W = \langle s_\alpha \mid \alpha \in \Delta \rangle$ (finite reflection group)
- Finite reflection group ⇒ finite Coxeter group

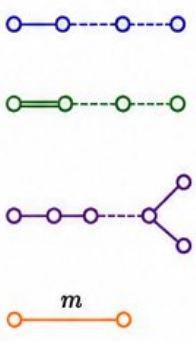
Key observations (in the example):

- 8 roots, 4 reflecting lines (hyperplanes)
- $|W| = 8$, and W acts on R^2

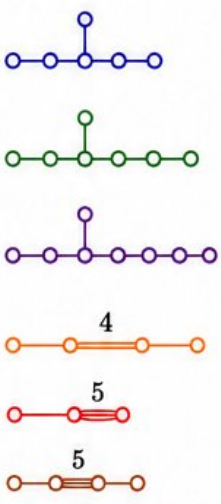
4. ALL COXETER GRAPHS – COMPLETE CLASSIFICATION

Edge label m = order of $s_i s_j$.
No edge means $m = 2$.
Double edge means $m = 4$.
Triple edge means $m = 6$.

INFINITE FAMILIES



EXCEPTIONAL TYPES



Key Fact & Classification Strategy

Coxeter graph of any finite reflection group is always positive definite.
Group of disconnected graph = direct product of groups of connected components
⇒ Focus on connected graphs.
The complete list above is exactly all connected positive definite Coxeter graphs.

CONCLUSION

Finite reflection groups in Euclidean space are completely classified by their Coxeter graphs.
The classification reduces to finding all connected positive definite Coxeter graphs — giving exactly the types
A_n, B_n, D_n, $I_2(m)$, E_6, E_7, E_8, F_4, H_3, H_4.

[1] J.E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Univ. Press, 1990.
[2] N. Bourbaki, *Groupes et algèbres de Lie*, Chapters 4–6, Hermann, Paris, 1968.